

Group	ID	Name	Topic	Dataset
Group 01	2102009	MD. MAHIBULLAH HOWLADER	Gradient Descent on linear regression Predict house prices	Kaggle - House Prices
	2102069	MD. ABDUL HAI FAHIM		
Group 02	2102027	MD. AFRIDI ALOM PRANTO	Gradient Descent for logistic regression - Classify breast cancer	UCI - Breast Cancer Wisconsin
	2102066	NAZMUS SAKIB		
Group 03	2102003	MD. ABU TAHER	Stochastic Gradient Descent (SGD) for text classification (spam/ham)	Kaggle - SMS Spam Collection
	2102007	MEHEDI HASAN		
Group 04	2102024	MD. SHARAFAT KARIM	Mini-batch SGD for digit recognition	Scikit-learn - Digits
	2102043	MD. TANVIR ISLAM		
Group 05	2102023	MD. SENARUL ISLAM	Comparison of Batch vs Stochastic Gradient Descent Boston housing	Scikit-learn - Boston Housing
	2102049	PROSENJIT MONDOL		
Group 06	2102051	MD. SAFIULLAH FARAJY	Gradient Descent with different learning rates	UCI - Student Performance
	2102036	ABUL BASAR		
Group 07	2102026	NAYEMA FERDOUSHI	Newton's Method for finding minimum of a cost function (regression)	UCI - Concrete Compressive Stren
	2102062	MAHAJABIN AFROZ MOUSHE		
Group 08	2102014	SADIA HOMAYRA NOWSHIN	Newton's Method vs Gradient Descent on logistic regression	Kaggle - Titanic Survival Data
	2102018	JANNATUL MIM		
Group 09	2102073	RAGHIB AL SHAHRIAR	Conjugate Gradient Method for solving linear regression	UCI - Energy Efficiency
	2102068	MD.NAZMUL HASAN KHOKUN		
Group 10	2102067	MOHAMMAD FARHADUL HAQUE FUAD	Conjugate Gradient on a large sparse matrix (image dataset)	Scikit-learn - Olivetti Faces
	2102056	SANDIPTA SAHA		
Group 11	2102038	MD.NOUSHAD BHUIYAN	L-BFGS optimization in classification model (Iris dataset)	Scikit-learn - Iris
	2102046	MD HAJEK ANJUM BENOY		
Group 12	2102010	NUSRAT JAHAN MIM	Comparison of L-BFGS and SGD in logistic regression	Kaggle - Heart Disease UCI
	2102063	SANZIDA ISLAM NUHA		
Group 13	2102002	RUSNI AKTER	LU Decomposition to solve linear equations in ML	Synthetic Linear Systems
	2102006	UDITA SARKAR CHANDRABINDU		

Group 14	2102012	AISHWARIYA SARDER	LU Decomposition in image compression	Any grayscale image - use OpenCV
	2102075	AFRIN JAHAN		
Group 15	2102047	PUSPITA BAIDYA	QR Decomposition in linear regression	UCI - Wine Quality
	2102055	MST. MEHERIN JAHAN JUI		
Group 16	2102033	AKASH PAUL	QR vs LU for solving ML least squares	UCI - Airfoil Self-Noise
	2102080	KLINTON CHAKMA		
Group 17	2102048	MD. MEHEDI HASAN MONIR	Trapezoidal Rule to approximate loss function	Synthetic quadratic loss on dummy
	2102050	NUR MOHAMMAD NAIM		
Group 18	2102020	MD. SADMAN KABIR BHUIYAN	Trapezoidal Rule in calculating area under ROC Curve	UCI - Diabetes
	2002060	RAJESH BISWAS		
Group 19	2102013	MAHEDI HASAN NAZMUL	Compare Trapezoidal and Simpson's Rule in model evaluation	Kaggle - Bank Marketing
	2102077	TANJIL ISLAM		
Group 20	2102034	MD. AL MAMUN	Using Trapezoidal Rule in numeric approximation of AUC-PR	UCI - Hepatitis C
	2102042	MD. NAIMUR RAHMAN NAIEM		
Group 21	2102008	ABDUL JOBAYER	Bisection Method to find root of ML loss gradient	Synthetic logistic regression loss fu
	2102052	MOHAMMED SAKIB HASAN		
Group 22	2102057	SHAID IBNA SOBHAN	Bisection Method for threshold tuning (binary classification)	UCI - Adult Income
	2102072	MD. SAFAYET HOSSAIN		
Group 23	2102037	MD. SABBIR KHAN MOYEEM	Newton-Raphson Method for solving logistic regression weights	UCI - Parkinson's Disease
	2102059	MD. ABDUL KAIYUM		
Group 24	2102015	MD. RIADUL ISLAM MAHI	Comparison of Newton-Raphson and Gradient Descent	Kaggle - Insurance Claim Prediction
	2102070	JAHID HASAN		
Group 25	2002014	SANJANA TASNIM PROME	Root-Finding in Neural Network Loss Function	Kaggle - Fashion MNIST
	2102076	JANNATUL AKIBA		
Group 26	2102017	DURJOY DAS	Use LU Decomposition + Gradient Descent hybrid method	UCI - Forest Fires
	2102060	HADIBUZZAMAN		

Group 27	2102016	MD. MEHEDI HASAN	Apply QR Decomposition in PCA preprocessing	Scikit-learn - Digits
	2102074	ABDULLAH		
Group 28	2102054	AMIT KUMAR BARMAN	PCA with L-BFGS optimization	Kaggle - Face Images
	1902047	SAAD ABU SAMI		
Group 29	2102030	YASIN ARAFAT	Trapezoidal Rule to numerically integrate KL-divergence	Any NLP dataset (e.g., Kaggle - Ne
	2102021	SADMAN HAFIZ SHUVO		
Group 30	2102064	MD. MOBIN HAQUE	Newton's Method on Cost Function Minimization (custom)	Synthetic regression dataset
	2002079	MD. TOFAIEL HUSSAIN TOTA		
Group 31	2102040	MAHIR ASHAB	Predict fuel efficiency using SGD	UCI - Auto MPG
	2102078	HRIVU SAMADDAR		
Group 32	2102071	JIHADUL ISLAM	Optimize portfolio risk using Newton's method	Yahoo Finance - historical stock pri
	2102058	MD. SIFAT		
Group 33	2102041	OMAR FARUK TANVIR	Apply QR to recommend movies via matrix factorization	Kaggle - MovieLens
	2102039	SUVO BISWAS		
Group 34	2102032	ARIFUL ISLAM MASUM	Bisection method to estimate optimal learning rate	Any classification dataset
	2102065	IMAMUL KABIR ANAN		
Group 35	2102005	MD. KAMRUZZAMAN	LU for solving kernel matrix in SVM	Scikit-learn - SVM example dataset
	2002082	ABHINASH KUMAR SAH		
Group 36	2102022	NOSHIN NAZIA	Compare SGD vs L-BFGS in spam detection	Kaggle - Email Spam Dataset
	2102035	BASUDHA SHOME		
Group 37	2002005	MD. TASNIM FERDOUS	Newton-Raphson in time series forecasting	Kaggle - Air Quality Data
	2002022	ARUP GHOSH		
Group 38	2002049	MD. MAYNUL ISLAM	LU/QR in solving system of equations from linear regression	Scikit-learn - California Housing
	2002066	ASHFIKUR RAHMAN		
Group 39	2102019	HRIDOY CHANDRA SARKER	SGD for recommender system matrix factorization	Kaggle - Anime Recommender
	2102025	SEEMANTA SHILL		

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